Experiment No: 10

Title: Implement UART communication using Data Structures in C

Aim : To implement UART (Universal Asynchronous Receiver/Transmitter) communication using data structures in C.

Mapped to Course Outcome:

Theory:

UART (Universal Asynchronous Receiver Transmitter) is a serial communication protocol used to transmit and receive data byte by byte.

In software, data structures like queues or circular buffers are used to manage incoming and outgoing data efficiently.

• Transmit Buffer: Stores data before sending over UART.

• Receive Buffer: Stores received data until it is read by the application.

A circular queue is preferred as it optimizes memory and supports continuous buffering.

Algorithm:

1. Initialize UART hardware configuration (baud rate, parity, stop bits).
2. Implement circular buffer using an array and head/tail pointers.
3. On transmit:
   • Enqueue data to TX buffer.
   • Dequeue from buffer and send byte over UART.
4. On receive interrupt:
   • Read byte from UART register.
   • Enqueue to RX buffer.
5. Provide interface functions: uart_send(), uart_receive(), is_buffer_full(), is_buffer_empty().

Code:

```
#include<stdio.h>
#define MAX 5

char tx[MAX],rx[MAX];
int front=-1,rear=-1;

int isFull(int f,int r){
   return (r+1)%MAX==f;
}

int isEmpty(int f){
   return f==-1;
}
```

```
void enqueue(char q[],int *f,int *r,char x){
   if(isFull(*f,*r)){
      printf("Buffer Full\n");
      return;
   }
   if(*f==-1)
      *f=0;
   *r=(*r+1)%MAX;
   q[*r]=x;
}

char dequeue(char q[],int *f,int *r){
   char x;
   if(isEmpty(*f))
      return '\0';

   x=q[*f];
   if(*f==*r)
      *f=*r=-1;
   else
      *f=(*f+1)%MAX;

   return x;
}

void uart_send(char x){
   enqueue(tx,&front,&rear,x);
}

void uart_receive(char x){
   enqueue(rx,&front,&rear,x);
}

int main(){
   int i;
   char x;

   printf("Enter data to transmit: ");
   for(i=0;i<3;i++){
      scanf(" %c",&x);
      uart_send(x);
   }

   printf("Transmitted data: ");
   while(!isEmpty(front)){
      printf("%c ",dequeue(tx,&front,&rear));
   }
```

```
    front=rear=-1;

    enqueue(rx,&front,&rear,'A');
    enqueue(rx,&front,&rear,'B');
    enqueue(rx,&front,&rear,'C');

    printf("\nReceived data: ");
    while(!isEmpty(front)){
        printf("%c ",dequeue(rx,&front,&rear));
    }

    return 0;
}
```

Output:

```
Enter data to transmit: H E L
Transmitted data: H E L
Received data: A B C
```

M.M.C. O. E.
Page No: Date:

Assignment 10

Q1 What is UART.
-> A serial communicatn protocol used to send & receive data b/w devices

Q2 Why use circular buffer for UART ?
-> It optimizes space and allows continuous data flow without memory shifting

Q3 What happens when buffer is full?
-> new data is lost or should be managed until space is available.

Q4 Is UART synchronous?
-> UART is Asynchronous?

Q5 What is baud rate?
-> The speed of data transfer, measured in bits per second (bps).